Politecnico di Milano

School of Industrial and Information Engineering
M.Sc. in Aeronautical Engineering

ASSIGNMENT 2

Structural Dynamics and Aeroelasticity

Dynamic Aeroelastic Analysis of a Strut-Braced Wing with Tip Tank

Matteo Marwan Graziani - 307177
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1 Numerical Result Report

1.1 Free-Vibration Convergence Analysis (Point 1.a)

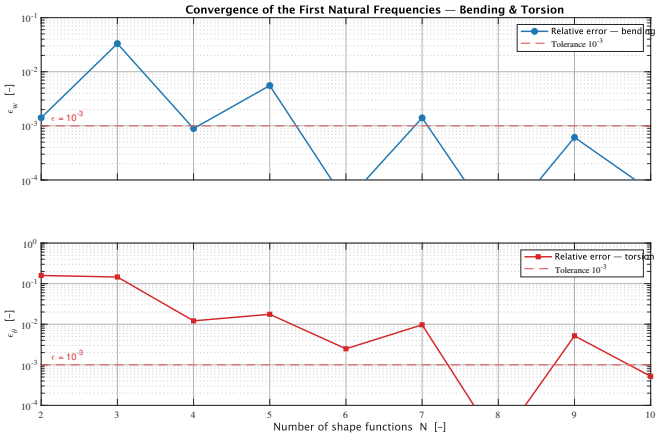


Figura 1: Ritz-Galerkin convergence for the strut-braced wing fundamental modes: bending relative error ϵ_w (top) and torsion relative error ϵ_θ (bottom) vs. number of shape functions N (tolerance $= 10^{-3}$).

The Ritz-Galerkin convergence profiles for the uncoupled natural frequencies are shown in Fig. 1. The bending error (ϵ_w) oscillates due to the shear discontinuity from the midspan strut ($y = b/2$), while the torsional error (ϵ_θ) decays slower due to the tip tank mass (M_T, J_T).

The modes converge to $\omega_{w1} = 15.73$ rad/s and $\omega_{\theta 1} = 35.16$ rad/s. The $\epsilon < 10^{-3}$ tolerance is met at $N_w = 4$ (bending) and $N_\theta = 8$ (torsion). These minimal orders are selected for subsequent analyses to guarantee the required accuracy while preventing numerical ill-conditioning and MATLAB solver crashes.

1.2 Frequency Response Function (Point 1.b)

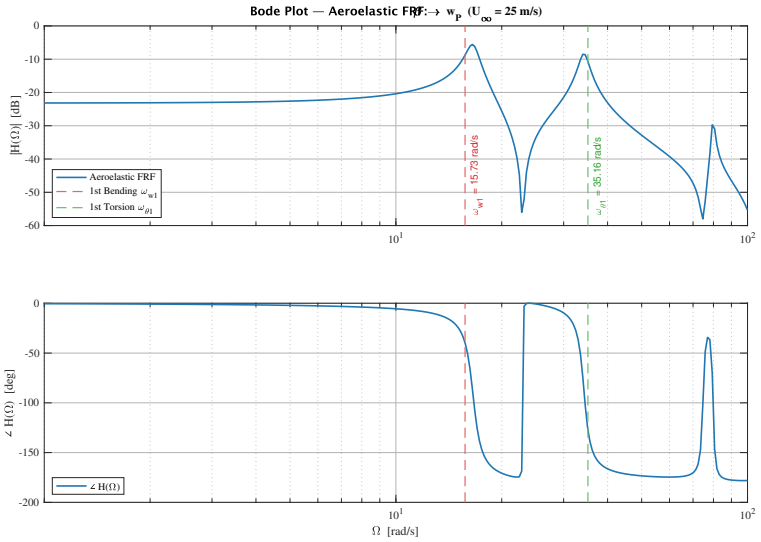


Figura 2: Aeroelastic Frequency Response Function (FRF) magnitude (top) and phase (bottom) from harmonic aileron deflection $\beta(t)$ to tip displacement $w_P(t)$ at $U_\infty = 25$ m/s.

The aeroelastic Bode plot in Fig. 2 depicts the dynamic response of point P (on the leading edge at $y = b$) under a quasi-steady aerodynamic regime ($U_\infty = 25$ m/s). The magnitude plot clearly highlights two dominant resonance peaks corresponding to the fundamental aeroelastic modes: the first bending mode at $\omega_{w1} = 15.73$ rad/s and the first torsional mode at $\omega_{\theta 1} = 35.16$ rad/s. The aerodynamic coupling introduces visible non-collocated effects, evidenced by the sharp anti-resonance (transmission zero) located between the two modal frequencies, accompanied by a sudden phase jump. At higher frequencies (≈ 85 rad/s), a third low-amplitude peak emerges, indicating that the selected shape functions properly retain high-frequency structural dynamics without compromising numerical stability.

1.3 Flutter Analysis (Point 1.c)

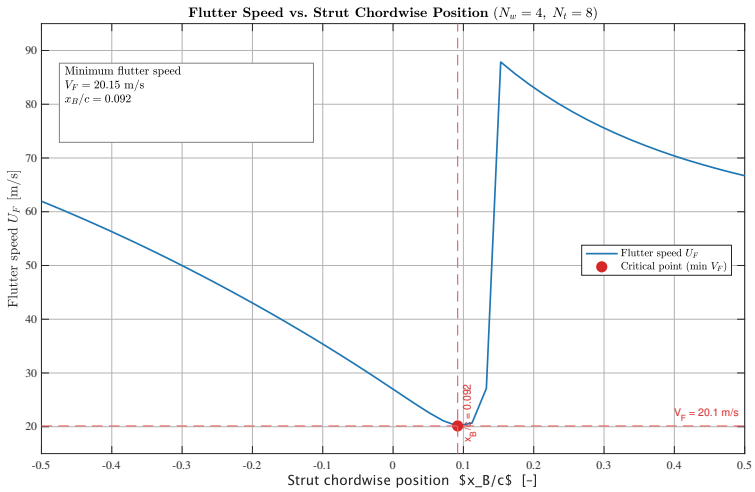


Figura 3: Aeroelastic flutter speed U_F as a function of the non-dimensional strut chordwise attachment position x_B/c at midspan ($y = b/2$).

The sensitivity of the flutter speed to the chordwise location of the strut attachment point B is shown in Fig. 3. The structural stability boundary undergoes significant variations depending on how the elastic axis is artificially stiffened by the strut. A distinct minimum flutter speed of $V_F = 20.15$ m/s is reached at $x_B/c = 0.092$, where the effective elastic center of the braced section shifts close to the aerodynamic center ($x_{AC}/c = -0.25$) or the center of gravity ($x_{CG}/c = 0.15$), maximizing the adverse pitch-plunge aeroelastic coupling. Moving the attachment point further aft ($x_B/c > 0.1$) triggers a sharp discontinuity and a localized increase in U_F, indicating a structural mode switch where the critical instability shifts from a classical bending-torsion coupled flutter to a higher-frequency single-degree-of-freedom torsional mechanism.

1.4 Step Gust – Direct Recovery of Root Bending Moment (Point 1.d)

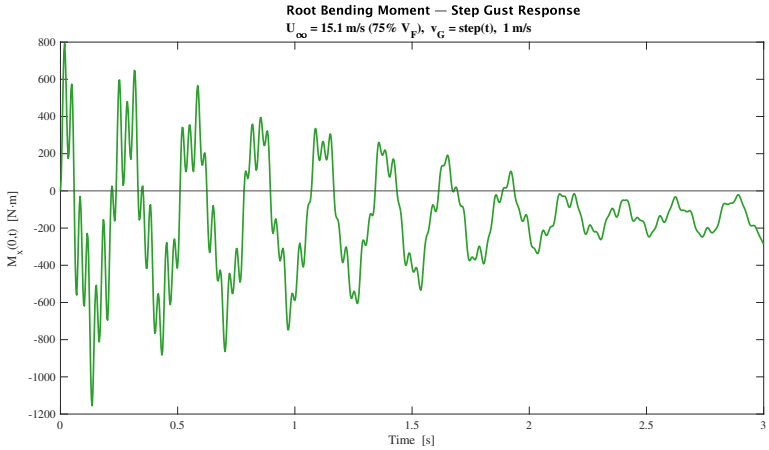


Figura 4: Transient root bending moment $M_x(0,t)$ response to a 1 m/s vertical step gust at $U_\infty = 15.1$ m/s $(75\%V_F)$ with direct force recovery.

The transient response of the root bending moment due to an instantaneous vertical step gust is presented in Fig. 4. The sharp onset of the gust induces an immediate aerodynamic load increment, triggering a high-amplitude multi-frequency structural vibration. The response is dominated by two main coupled mechanisms: a high-frequency component that modulates the peak values, and a lower-frequency carrier wave corresponding to the fundamental bending mode. Because the operational flight speed is close to the stability boundary $(75\%V_F)$, the aerodynamic damping is relatively low, leading to a slow decay rate of the oscillations. The system asymptotically settles toward a non-zero, negative steady-state bending moment (≈ -200 Nm), which represents the pure static aeroelastic equilibrium under the sustained gust velocity field.

1.5 Stochastic White-Noise Gust – Variance via Lyapunov (Point 1.e)

The stochastic response of the root bending moment $M_x(0,t)$ under a continuous, uniform vertical white-noise gust of unit intensity is evaluated by solving the corresponding continuous-time Lyapunov state equation. Operating at a flight speed of $U_\infty = 15.1$ m/s $(75\%V_F)$, the system exhibits a root bending moment variance of $\sigma^2 = 1.0403 \times 10^9$ $(\mathrm{N}\cdot\mathrm{m})^2$, yielding a standard deviation of $\sigma = 3.2253 \times 10^4$ $\mathrm{N}\cdot\mathrm{m}$.

This elevated statistical variance is directly attributable to the low aeroelastic damping margins inherent to operations near the flutter boundary. Because the subcritical aerodynamic damping decreases as the velocity approaches V_F, the structural system acts as a highly selective, narrow-band filter. Consequently, the low-damped fundamental bending and torsional modes resonantly amplify the white-noise power spectral density across their respective natural frequencies, leading to large-scale, random structural excursions in the root loads.